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Binomial Regression Inference

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The maximum likelihood estimator is unbiased.

Due May 26, 11:59 PM IST

3 / 3 pointsTry again

- ☐ True
- ☒ False
- ☒ Correct

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2. As the sample size n tends to infinity, the distribution of the maximum likelihood estimator becomes $\hat{\theta}_{ML} \sim N(\theta, I^{-1}(\theta))$

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 3 / 3 points

- ☒ True
- ☐ False
- ☒ Correct

3. Let β_j be the parameter associated with predictor x_j in a binomial regression model. For a reasonably large sample size n , a standard normal "z-test" can be used to test whether x_j should be in the model. 3 / 3 points

- ☒ True
- ☐ False
- ☒ Correct

4. Let $X_1, \dots, X_n$ be a random sample from a distribution with pdf $f(x; \theta)$, and let $\hat{\theta}$ be the maximum likelihood estimator of θ . Then 3 / 3 points

$$\left(\hat{\theta}_{ML} - \frac{1.96}{\sqrt{nI(\theta)}}, \hat{\theta}_{ML} + \frac{1.96}{\sqrt{nI(\theta)}} \right)$$

is an approximate 95% confidence interval for θ .

- ☒ True
- ☐ False
- ☒ Correct

5. Goodness of fit metrics - such as the residual deviance - are only useful for the binomial regression with a relatively large number of trials (e.g., $n > 5$). 3 / 3 points

- ☒ True
- ☐ False
- ☒ Correct

6. Consider a logistic regression fit an independent response $Y_i \sim \text{Binomial}(1, p)$ and a single predictor variable x . The linear predictor is: 4 / 4 points

$$\eta_i = \beta_0 + \beta_1 x_i$$

Test $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$ by computing the appropriate p-value, rounded to the hundredths place.

Coefficients:

	Estimate	Std. Error
(Intercept)	-1.2467	0.6347
x	1.4224	1.1541

- ☒ 0.22
- ☐ 1
- ☐ 0.05
- ☐ 0.11

Correct

You can estimate the t-test (t-stat) as:

$$t_stat = \text{Estimate} / \text{SD Error}$$

Computing p-value when t_stat or z-score is given:

$$p_value = 2 * (1 - \text{pnorm}(\text{abs}(t_stat)))$$

7. Consider a logistic regression fit an independent response $Y_i \sim \text{Binomial}(1, p)$ and a single predictor variable x . The linear predictor is: 4 / 4 points

$$\eta_i = \beta_0 + \beta_1 x_i$$

Use maximum likelihood theory to construct an approximate 95% confidence interval for β_0 . Round all values to the hundredths place.

Coefficients:

	Estimate	Std. Error
(Intercept)	-1.2467	0.6347
x	1.4224	1.1541

- ☒ (-2.48, -0.02)
- ☐ (-0.47, 3.31)
- ☐ (-2.28, -0.22)

Correct